🧊 The Melting Race! 🧊

Physical Science — Sunshine & Warmth — Unit 2, Lesson 4 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth
🔢 **Lesson:** 4 of 6

Three ice cubes, three conditions — who will melt first?

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate the Melting Race, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Sunlight can cause solids to melt by warming them. But **where** the ice is placed and **what color** it sits on makes a huge difference in how fast it melts! In this experiment, we put ice cubes in three different conditions and have a melting race. The results connect everything we've learned so far: dark colors absorb more heat, direct sunlight warms more than shade, and reflective surfaces can protect against warming.

🔬 The Science

Ice melts when enough heat energy reaches it to change it from solid water (ice) to liquid water. Sunlight delivers that heat energy. Dark plates absorb more sunlight and transfer that heat to the ice beneath — melting it faster. White plates reflect more sunlight, so less heat reaches the ice. Shade blocks direct sunlight almost entirely, so very little heat reaches the ice that way. In the extension, students test **insulation** — materials that slow down the transfer of heat. Aluminum foil reflects sunlight; dark cloth absorbs it; white paper is in between. This connects directly to Lessons 5 & 6 (designing a sun shelter).

> 💡 **Key vocabulary:** melt, solid, liquid, heat energy, absorb, reflect, insulate, shade, sunlight, predict

📦 Materials

🏠 What You Need

- 3 ice cubes (same size — use an ice cube tray)
- 1 dark plate or dark paper plate
- 1 white plate or light-colored dish
- Access to a sunny spot + a shaded spot nearby
- Timer
- **For the extension:** 3 more ice cubes + aluminum foil + small piece of dark cloth + white paper
- Student worksheet (included)

🔬 Let's Get Started

Step 1 — Make Your Predictions (5 min)
Show kids the three conditions:

- 🖤 Dark plate in the sun
- 🤍 White plate in the sun
- 🌑 Dark plate in the shade

Ask: *"Which ice cube do you think will melt the fastest? Which will melt the slowest? Why?"*
Have them circle their predictions on the worksheet.

Step 2 — Start the Race! (15–20 min)
Place one ice cube on each: dark plate in sun, white plate in sun, dark plate in shade.
Start the timer! Check every 5 minutes and note how much each cube has melted.
Kids can draw or describe what they see on their worksheet.

💡 **Expected winner:** Dark plate in the sun melts first! The dark plate absorbs heat AND the sun provides direct energy. The white plate in the sun is second (sun helps but plate reflects). The shaded dark plate is last — no direct sun, so melting is much slower.

Step 3 — Discuss the Results (5 min)
After the race, compare results to predictions. Were they right? Surprised?
Ask: *"Why did the dark plate in the sun win? What two things helped it melt fastest?"* (Dark color absorbed heat + direct sunlight provided heat energy.)

Step 4 — Extension: Insulator Challenge (10 min, optional)
Now try protecting ice cubes from melting! Wrap 3 more ice cubes in:
🪩 Aluminum foil | 🖤 Dark cloth | 🤍 White paper
Put all three in direct sun. Which wrapping keeps the ice cold longest?
(Foil reflects sunlight best, so it protects the ice most effectively in direct sun.)

💬 Discussion Question

🗣️ **"Polar ice caps are white and reflect sunlight. What would happen if they melted and were replaced by dark ocean water?"**

> *→ This is actually happening! White ice reflects most of the sunlight that hits it — like a giant mirror in the sky. But dark ocean water absorbs much more sunlight, warming the ocean further, which melts more ice. This creates a dangerous chain reaction called the "albedo feedback loop." The word "albedo" means how much light a surface reflects. White ice has high albedo (reflects lots); dark ocean has low albedo (absorbs lots). As ice melts and is replaced by dark water, Earth absorbs even more heat — speeding up warming further.*

🚀 Extensions

- 🌍 **Albedo Art:** Look up photos of the Arctic from satellite. White ice vs. blue ocean — you can see the difference in albedo right from space!
- 🏠 **Insulator Design:** Before Lesson 5, try to think: what's the BEST material to wrap an ice cube in to keep it from melting? Design a small test and report your findings.
- 🧊 **Kitchen Experiment:** Does ice melt faster in a metal bowl or a plastic bowl? (Metal conducts heat better!) Try it and see.
- 🌡️ **Time It:** Exactly how many minutes did each ice cube take to fully melt? Record the times and see how much faster the dark plate in sun was.

👩‍🏫 Parent/Teacher Notes

- **Ice cube size matters:** Use ice cubes from the same tray so they start the same size. Unequal starting sizes can skew the results.
- **Direct sun is key:** This experiment needs a genuinely sunny day. On a cloudy day, the differences will be much smaller. A sunny window works in a pinch.
- **Melting time:** Depending on temperature and sun intensity, full melting can take anywhere from 10 minutes to 45+ minutes. Just observe the progress rather than waiting for complete melting if time is short.
- **Albedo discussion:** The polar ice / albedo loop is real-world climate science. It's appropriate to explain this simply: "when the white ice disappears, the darker water absorbs more heat, which melts even more ice." You don't need to use the word "albedo" — the concept is what matters at this grade level.
- **Connection to Lessons 5–6:** The insulator extension directly previews the engineering challenge coming up — designing something to protect against heat. Great to ask: "How could we build something to protect our ice cube from the sun forever?"

🧊 Student Worksheet

Name: _____ Date: _____

🔮 My Predictions — Which will melt *first*?

Number the boxes 1 (melts first), 2 (second), 3 (melts last):

🖤

Dark Plate in the SUN ☀️

Write 1, 2, or 3

🤍

White Plate in the SUN ☀️

Write 1, 2, or 3

🌑

Dark Plate in the SHADE 🌳

Write 1, 2, or 3

👀 What I Saw — Draw the ice cubes after 10 minutes:

🖤

Dark Plate in Sun

🤍

White Plate in Sun

🌑

Dark Plate in Shade

🏆 The Winner!

The ice cube that melted first was on the: _____

It melted fastest because: _____

🛡️ Insulator Challenge — Which wrapping kept ice coldest?

🪩 🖤 🤍

Foil	Dark Cloth	White Paper
write or draw	write or draw	write or draw

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Expected Melting Order

1. 🥇 **Dark plate in the sun** — melts *first*. The dark plate absorbs sunlight AND direct sun adds more heat energy directly.
2. 🥈 **White plate in the sun** — melts SECOND. Direct sun still provides heat, but the white plate reflects some sunlight rather than absorbing it.
3. 🥉 **Dark plate in the shade** — melts LAST. No direct sunlight, so only ambient heat reaches the ice. Shade dramatically slows melting.

Insulator Results (Extension)

1. 🥇 **Aluminum foil** — best protection in sunlight! It reflects sunlight rather than absorbing it, keeping the ice underneath cooler. (Note: foil works here by *reflecting* radiant energy, not by insulating against conduction.)
2. 🥈 **White paper** — moderate; reflects some light, but also allows some heat through.
3. 🥉 **Dark cloth** — least protective in sunlight! It absorbs sunlight and transfers that heat directly to the ice — making it melt faster!

💡 Science Explanation for Parents

This experiment demonstrates that sunlight is the primary driver of melting here, and that both the **color of the surface** and **presence/absence of direct sunlight** matter. Aluminum foil works well against radiant heat (sunlight) because it reflects it rather than absorbing it. This is the same reason emergency (Mylar) blankets are silver — they reflect radiant warmth back to the body. Note: foil's effectiveness here is about *reflection*, not thermal insulation (which refers to slowing conductive or convective heat transfer). The albedo feedback concept previews real climate dynamics: as Arctic sea ice melts and is replaced by dark ocean water, the ocean absorbs more sunlight, warming further and melting more ice — a self-reinforcing feedback loop that climate scientists study intensively.

✅ What Success Looks Like

- Child correctly predicts or identifies that dark plate in sun melts ice fastest
- Child can explain: "dark colors absorb more sun, so they get warmer and melt ice faster"
- Child can describe that shade keeps ice colder by blocking sunlight
- Bonus: Child understands that foil keeps ice cold by reflecting sunlight away

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